



Reflexive AI

REFLEXIVE AI RESEARCH INITIATIVE

Human-Centered AI

Overview of relevant insights from scientific research

10 minutes

details & sources @ <https://github.com/LucidInstitute/Reflexive-AI-Initiative/tree/main/Literature>

DEFINING AI

AI can take over human thinking, and technology is never neutral.

// KEY FINDINGS

- 1 OECD definition.** A machine-based system that infers from input how to generate predictions, content, recommendations or decisions (OECD, 2024). More simply: “Computers that do the sorts of things minds can do” (Boden, 2016).
- 2 Minds, delegated.** Unlike earlier revolutions, thinking and creativity can now be delegated, not only physical action. (Boden, 2016; Hilbert, 2020). Robotics will also replace more physical action.
- 3 Technology embodies power.** It shapes how we see the world and imposes behaviors on people. (Winner, 1985; Latour, 1992)
- 4 Bias carries over.** AI can perpetuate biases already present in its training data. (Barocas & Selbst, 2016; Noble, 2018)

Individual gains are real, but they do not automatically become team gains.

// KEY FINDINGS

1

Firm level. Positive productivity effects across sectors, stronger when AI is integrated into work activities. (Necula et al., 2024; Czarnitzki et al., 2023)

2

Employee level. In a survey of 359 bank employees, AI use raised productivity and engagement. (Gusti et al., 2024)

3

Perception gap. Developers felt more productive, yet commit activity did not change and experienced developers were not faster. (Stray et al., 2026; Becker et al., 2025)

4

Lost downstream. Local gains are absorbed by workflow and organizational constraints. (DORA, 2025; Russo, 2026)

Whether AI helps or harms depends on trust, fit and culture.

// KEY FINDINGS

1

Engagement. AI raised engagement in two manufacturers, unless self-efficacy is low or jobs were cut. (Tortorella et al., 2025)

2

Trust. Low employer trust sharply reduces engagement; AI may bring an “alienational” contract. (Braganza et al., 2021)

3

Preconditions. Task fit, transparency and expertise. Black boxes breed aversion; bureaucracy builds “algorithmic cages”. (Bankins et al., 2023; Meijer et al., 2021)

4

Six displacements. Software engineers report epistemic, identity, normative, phenomenological, social and meaning shifts. (Tavantzis et al., in press)

AI can amplify existing inequality unless we design purposefully against it.

// KEY FINDINGS

- 1 Bias in practice.** A widely used health system under-recommended treatment for black relative to white patients. (Obermeyer et al., 2019)
- 2 Hard to remove.** Mitigation strategies exist, but implicit biases are difficult to eradicate. (Nazer et al., 2023; Howard & Borenstein, 2017)
- 3 Digital divide.** Inequality can grow between households that can use AI and those that cannot. (Cornelli et al., 2023)
- 4 Labor market.** Automation can replace low-skilled tasks, leaving some workers worse off despite seeming economic growth. (Agrawal et al., 2019; Li & Nichols, 2026)

Human-centered AI balances human control and automation, guided by shared values.

// KEY FINDINGS

- 1 One scale is not enough.** Early models run from no automation to full autonomy, as in self-driving cars, and force a choice. Too much control or too much automation can both add risk. (Sheridan & Verplank, 1978; SAE, 2014)
- 2 Two dimensions.** Human control and computer automation are separate axes. Life-critical systems such as pacemakers and self-driving cars need both high to be reliable, safe and trustworthy. (Shneiderman, 2020)
- 3 Copenhagen Manifesto.** 35 scholars set five values and twelve principles on human dignity, autonomy and collective wellbeing, including human sovereignty and responsible management. (Russo et al., 2024)
- 4 Working definition.** No agreed definition yet. Our draft: efficient work while people stay in control, motivated, connected, and free to question, change or reject AI decisions. (Initiative draft)

Governance must turn principles into structures, processes and enforcement.

// WHERE TO FOCUS

- 1 Answer how, not just what.** Ethical guidelines say what, not how, and can lose to commercial interests. Define structures, processes and relational mechanisms, with enforcement. (Mittelstadt, 2019; Mäntymäki et al., 2022)
- 2 Cover who, what, when and how.** Define who governs, what is governed, when across the lifecycle, and how through policies, tools and models. Think system-level, not just algorithm ethics. (Batool et al., 2025; Lu et al., 2022)
- 3 Go beyond fairness and privacy.** Most solutions address only one or two principles. Build in the human-centered ones: control, autonomy, wellbeing and creativity. (Batool et al., 2025; Russo et al., 2024)
- 4 Make it actionable.** Use concrete patterns, fit them to culture, leadership and up-skilling, and test with Action Research in real organizations. (Lu et al., 2022; Tavantzis et al., 2027)

Sixteen starting themes for human-centered AI

DIRECTION & CONTROL

Purpose & Strategy ^{T-K}

What does human-centered mean here, and which real need does each AI use serve?

Human Agency ^{T-K}

Who decides, how far may AI agents act alone, and who answers for the result?

Governance & Ethics ^{T-K}

Which rules, roles and enforcement turn our principles into daily practice?

Review & Adaptation ^{T-K}

How do we track outcomes, harms and wellbeing, and adapt at a human pace?

PEOPLE & WORK

Meaningful Work ^K

How do ensure work remains fulfilling, motivating and purposeful to employees?

Work Design ^{T-K}

How do we split tasks between people and AI and redesign workflows?

Workforce & Learning ^{T-K}

Which skills must people build or keep to use, verify and question AI?

Quality & Oversight ^{T-K}

How do we verify AI output, surface uncertainty and prevent over-reliance?

CULTURE & VOICE

Communication ^{T-K}

Do people know where, how and why AI is used, and how their data is handled?

Participation ^{T-K}

How do employees and their representatives co-design and contest AI use?

Questioning Culture ^{T-K}

Can people experiment, make mistakes and reject AI output without pressure?

Power & Dynamics ^T

Whose interests, roles and timing shape AI, and where does resistance come from?

BOUNDARIES & DUTIES

Risk & Boundaries ^{T-K}

Where is AI restricted or off-limits, e.g. HR, health, monitoring of staff?

Legal & Compliance ^{T-K}

Which laws and regulations, like the EU AI Act, apply to each AI use?

IP & Data Ownership ^T

Who owns AI inputs, outputs and data, and what may leave the organization?

Sustainability ^K

What is the environmental and long-term societal cost of our AI use?

THANK YOU

**Now its up to us to create concrete
help for organizations**